



Question No	1	2	3	4	5	6	7	Total	Total
Mark	$\overline{12}$	$\overline{12}$	$\overline{12}$	$\overline{12}$	$\overline{12}$	$\overline{12}$	$\overline{12}$	$\overline{84}$	$\overline{100}$

Question 1 (12 marks)**Marks**

- (a) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$ 1
- (b) Differentiate $\log_e(\sin^3 x)$, writing the answer in simplified form. 2
- (c) Find the range of the function $f(x) = x^2 - 6x + 10$. 2
- (d) Find the acute angle between the lines $-x + 2y + 4 = 0$ and $3x + y + 1 = 0$. Give your answer to the nearest minute. 2
- (e) Evaluate $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}}$ 2
- (f) i) Show that $x-2$ is a factor of $x^3 - 4x^2 + 7x - 6$ 1
ii) Show why $x^3 - 4x^2 + 7x - 6 = 0$ has only 1 real root. 2

Question 2 (12 marks)

- (a) Find the exact value of $\cos \left[\sin^{-1} \left(\frac{1}{3} \right) \right]$ 2
- (b) Solve the equation $3x^3 - 17x^2 - 8x + 12 = 0$ given that the product of two of the roots is 4. 3
- (c)
- (i) Express $\sqrt{12} \sin x + 2 \cos x$ in the form $R \cos (x - \alpha)$ where $R > 0$ and $0 \leq x \leq \frac{\pi}{2}$ 2
- (ii) Hence solve $\sqrt{12} \sin x + 2 \cos x = -3$ for $0 \leq x \leq 2\pi$ 2
- (d) Solve the inequality $\frac{x^2 - 9}{x} \leq 8$ 3

Question 3 (12 marks)

- (a) (i) Show why the equation $\ln(x+1) + x - 1 = 0$ must have a root between $x=0$ and $x=1$. 1
- (ii) Given that the solution of $\ln(x+1) + x - 1 = 0$ is approximately equal to $\frac{1}{2}$, use one application of Newton's method to get a better approximation. Write your answer correct to two decimal places. 2
- (b) The acceleration of a particle P is given by $a = 32x(x^2 + 1)$ where x cm is the displacement at time t sec. Initially P starts from the origin with velocity 4 cm/s.
- (i) Use the fact that $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ to show that $v = \pm 4(x^2 + 1)$ 2
- (ii) Justify why the positive solution is the correct one. 1
- (iii) Find x in terms of t . 2
- (c) Prove using mathematical induction that, for all positive integers $n \geq 1$,

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3) \times (4n+1)} = \frac{n}{4n+1} \quad 4$$

Question 4 (12 marks)

- (a) Find $\int \cos^2 3x \, dx$ 2
- (b) (i) Show that the maximum value of $2x(1-x)$ is $\frac{1}{2}$ 1
- (ii) Find the range of the function $f(x) = \sin^{-1}\{2x(1-x)\}$, with domain $0 \leq x \leq 1$. 2
- (c) (i) Show that $T = 22 + Ae^{kt}$ is a solution to the equation $\frac{dT}{dt} = k(T - 22)$. 1
- (ii) A wealthy industrialist is found murdered in his home. When police arrived on the scene at 11:00 pm, the temperature of the body was 31°C , and one hour later it was 30°C . The temperature of the room where the body was found was 22°C . Using Newton's law of cooling $\frac{dT}{dt} = k(T - 22)$, and the fact that normal body temperature is 37°C , estimate the time that the murder occurred. 3

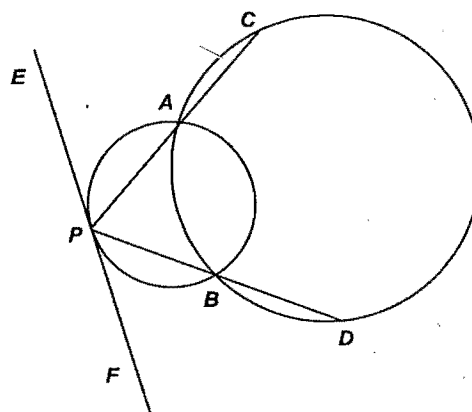
[Question 4 continued]

- (d) Copy the diagram into
your answer booklet.

PAC, PBD are straight lines.

EF is the tangent at P .

Prove $CD \parallel EF$



3

Question 5 (12 marks)

- (a) Use the substitution $u = x - 1$ to evaluate $\int_2^4 \frac{x}{(x-1)^2} dx$ 3

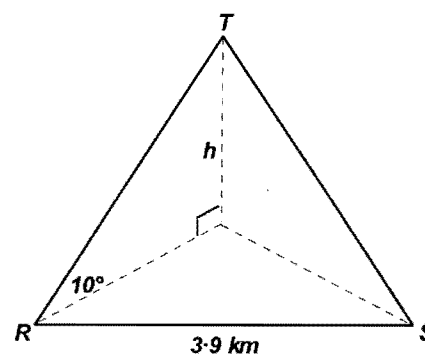
- (b) A particle moves along a straight line in such a way that its distance x cm from a fixed point O at time t seconds is $x = 2 \cos 3t$.

- (i) Show that the particle is moving in Simple Harmonic Motion. 2
(ii) Write down the period of the motion. 1
(iii) Find the particle's speed when it is first 1 cm from O . 2

- (c) Two boats, *Rascal* and *Sirocco*, are sailing near Ball's Pyramid, a giant pillar of rock that rises from the sea. 4

The boats are 3.9 km apart, and the angle of elevation of the top T of Ball's Pyramid from *Rascal* is 10° .

Given that $\angle TRS$ and $\angle TSR$ are 65° and 48° respectively, find the height h of Ball's Pyramid to the nearest metre.



Question 6 (12 marks)

(a) Water is being pumped into an empty inverted conical tank – that is, a tank with its apex down – at a rate of $300\,000\text{ cm}^3/\text{min}$. The tank is 6 metres high and the diameter at the top is 4 metres. Let h be the height of the water and r the radius of the water surface at time t minutes.

Calculate the rate at which the water level is rising at the instant the water level reaches 2 metres. (Write your answer to 1 decimal place). 4

(b) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$, which has focus S .

(i) Derive the equation of the tangent to the parabola at P . Hence show that the tangent meets the x-axis at the point $T(ap, 0)$. 3

(ii) Find the coordinates of M , the point that divides ST externally in the ratio $2 : 1$ 2

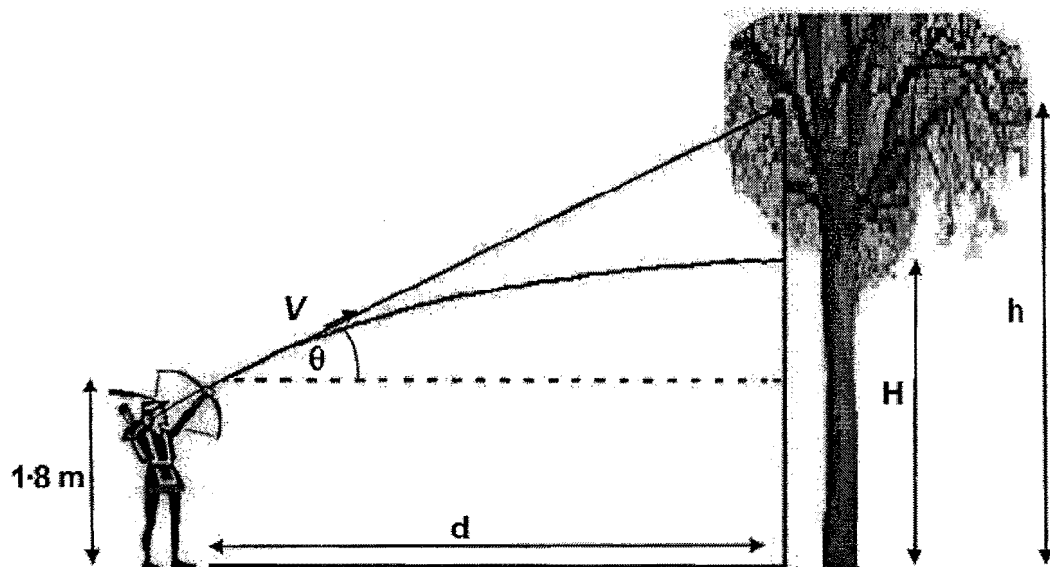
(iii) Describe geometrically the locus of M as P moves on the parabola. 1

(c) A function is given by the rule $f(x) = \log_e \frac{1+x}{1-x}$. Find the rule for the inverse function, $f^{-1}(x)$. 2

Question 7 (12 marks)

(a) Robin Hood aims his arrow at an acorn which is on an oak tree d metres away. The acorn is h metres above the ground, and Robin releases the arrow from a point 1.8 metres above the ground, at an angle of elevation of θ degrees. Robin can vary the initial velocity V of the arrow.

At the instant Robin releases the arrow, the acorn begins to fall vertically downwards under gravity, with acceleration g . (Neglect air resistance).



[Question 7 continued]

- (i) Taking the ground at Robin's feet as the origin of coordinates, show that at time t the x - and y -coordinates of the arrow's tip are

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - \frac{gt^2}{2} + 1.8 \quad 3$$

- (ii) Show that, when the arrow reaches the tree, its vertical height above the ground is given by

$$H = d \tan \theta - \frac{gd^2 \sec^2 \theta}{2V^2} + 1.8 \quad 2$$

- (iii) Robin's arrow hits the acorn as it falls. Show mathematically why this will in fact always be the case in this situation, no matter what the initial velocity of the arrow.

3

(Provided, of course, that it is great enough to reach the tree!)

- (b) On a toss of two dice, John throws a total of 5. Find the probability that he will throw another 5 before he throws 7.

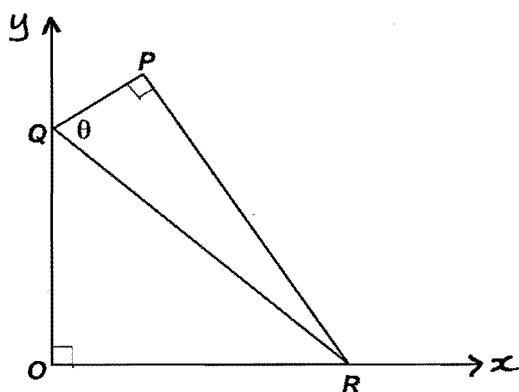
2

- (c) A triangle PQR , right angled at P and with $\angle PQR = \theta$, slides on a horizontal floor with its vertices Q and R in contact with perpendicular walls.

- (i) State why $OQPR$ is a cyclic quadrilateral.
- (ii) Derive the equation of the locus of P .

1

1



QUESTION 1

(a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = 3 \cdot \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$
 $= 3 \cdot 1$
 $= 3$

(b) $\frac{d}{dx} \ln(\sin^3 x) = \frac{3 \sin^2 x \cdot \cos x}{\sin^3 x}$
 $= \frac{3 \cos x}{\sin x}$
 $= 3 \cot x$

(c) $f(x) = x^2 - 6x + 10$
 $= (x-3)^2 + 1$

The graph of $y = f(x)$ is a concave up parabola, vertex (3,1). $\therefore R: y \geq 1$

Alternatively,

$f'(x) = 2x - 6$
 $= 0$ when $x = 3$.

Since $f''(x) = 2 > 0$ $\therefore$ minimum turning point at $x = 3, y = 1$. $\therefore R: y \geq 1$

(d) $m_1 = \frac{1}{2}, m_2 = -3$

$\therefore \tan \theta = \left| \frac{\frac{1}{2} - (-3)}{1 + \frac{1}{2}(-3)} \right|$

$= 7$

$\therefore \theta = 81^\circ 52'$ (nearest minute)

Q1 cont'd

(e) $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx = \left[\sin^{-1} \frac{x}{2} \right]_1^{\sqrt{3}}$
 $= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2}$
 $= \frac{\pi}{3} - \frac{\pi}{6}$
 $= \frac{\pi}{6}$

(f) (i) Let $x^3 - 4x^2 + 7x - 6 = P(x)$

Then $P(2) = 8 - 16 + 14 - 6 = 0$

$\therefore (x-2)$ is a factor of $P(x)$

(ii)
$$\begin{array}{r} x^2 - 2x + 3 \\ x-2 \overline{) x^3 - 4x^2 + 7x - 6} \\ \underline{x^3 - 2x^2} \\ -2x^2 + 7x \\ \underline{-2x^2 + 4x} \\ 3x - 6 \\ \underline{3x - 6} \\ 0 \end{array}$$

$\therefore P(x) = (x-2)(x^2 - 2x + 3)$

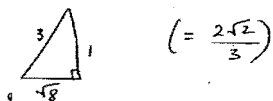
But $x^2 - 2x + 3$ has $\Delta = (-2)^2 - 4(1)(3) = -8 < 0$

and thus has no real roots.

$\therefore x = 2$ is the only real root.

QUESTION 2

(a) $\cos \left[\sin^{-1} \left(\frac{1}{3} \right) \right] = \frac{\sqrt{8}}{3}$



(b) $3x^3 - 17x^2 - 8x + 12 = 0$

Let roots be α, β, γ , and

Suppose $\alpha\beta = 4$

Now $\alpha\beta\gamma = -\frac{12}{3} = -4 \therefore \gamma = -1$

Also $\alpha + \beta + \gamma = \frac{17}{3}$

$\therefore \alpha + \frac{4}{\alpha} - 1 = \frac{17}{3}$ (since $\alpha\beta = 4$)

$\therefore 3\alpha^2 - 20\alpha + 12 = 0$

$(3\alpha - 2)(\alpha - 6) = 0$

$\therefore \alpha = \frac{2}{3}$ or 6

ie. roots are $-1, \frac{2}{3}, 6$.

(c) Let $\sqrt{12} \sin x + 2 \cos x = R \cos(x - \alpha)$

(i) $R \cos \alpha = 2$
 $R \sin \alpha = \sqrt{12}$

$\therefore R = 4$
 $\tan \alpha = \frac{\sqrt{12}}{2} = \sqrt{3} \therefore \alpha = \frac{\pi}{3}$

$\therefore$ expression $= 4 \cos(x - \frac{\pi}{3})$

(ii) $4 \cos(x - \frac{\pi}{3}) = -3, 0 \leq x \leq 2\pi$

$\therefore \cos(x - \frac{\pi}{3}) = -\frac{3}{4}$

2.864

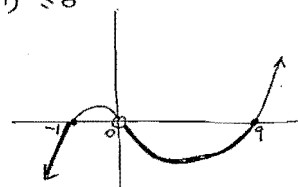
Q2 cont'd

(d) $\frac{x^2 - 9}{x} \leq 8$

$\therefore x(x^2 - 9) \leq 8x^2 \quad (x \neq 0)$

$\therefore x[x^2 - 9 - 8x] \leq 0$

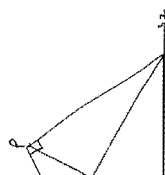
$x(x-9)(x+1) \leq 0$



$\therefore x \leq -1$ or $0 < x \leq 9$.

Q7 CONTINUED

(i) OQPR is a cyclic quadrilateral since opposite angles $\angle OPR$ and $\angle ORQ$ are supplementary (each $= 90^\circ$).



Join PO.
 From OQ, PR is a chord of a circle.
 So $\angle POQ = \angle POR$ (Angles in the same segment).
 But this is true for all positions of P as Q, R slide along the walls.
 $\therefore$ the gradient of PO is always $\tan \theta$.
 Since PO goes through the origin, P's locus is the straight line $y = (\tan \theta)x$.

QUESTION 3.

(a) (i) Let $f(x) = \ln(x+1) + x - 1$

Then $f(0) = \ln 1 + 0 - 1 = -1 < 0$

and $f(1) = \ln 2 + 1 - 1 = \ln 2 > 0$

$f_0(0)$ and $f(1)$ have opposite signs,
hence a root lies between 0 and 1.

(ii) $f'(x) = \frac{1}{x+1} + 1$

Let $x_1 = \frac{1}{2}$

Then $x_2 = \frac{1}{2} - \frac{f(\frac{1}{2})}{f'(\frac{1}{2})}$

$= \frac{1}{2} - \frac{(\ln 1.5 - 0.5)}{(\frac{1}{1.5} + 1)}$

≈ 0.5567209

$= 0.56$ (to 2 d.p.).

(b) (i) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 32x(x^2+1)$

$= 32x^3 + 32x$

$\therefore \frac{1}{2} v^2 = 8x^4 + 16x^2 + C$

But $x=0, v=4$

$\therefore \frac{1}{2} \cdot 4^2 = 0 + C \therefore C = 8$

$\therefore v^2 = 16x^4 + 32x^2 + 16$

$= 16(x^4 + 2x^2 + 1)$

$= 16(x^2+1)^2$

$\therefore v = \pm 4(x^2+1)$

Q3 cont'd

(b) (ii) At $t=0, x=0$ and $v=4$.

So particle is moving to the right. But since $a > 0$ when $x > 0$, it will keep moving to the right. $\therefore v > 0$

Thus $v = 4(x^2+1)$

(iii) $v = \frac{dx}{dt} = 4(x^2+1)$

$\therefore \frac{dt}{dx} = \frac{1}{4} \cdot \frac{1}{x^2+1}$

$t = \frac{1}{4} \tan^{-1} x + C$

But at $t=0, x=0 \therefore C=0$

$\therefore 4t = \tan^{-1} x$

$\therefore x = \tan 4t$

(c) We prove that, for $n \geq 1$,

$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$

When $n=1$

LHS $= \frac{1}{1 \times 5} = \frac{1}{5}$

RHS $= \frac{1}{4(1)+1} = \frac{1}{5}$

$\therefore$ it is true for $n=1$

Assume it is true for $n=k$ ($k \geq 1$)

$\therefore \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$ (*)

Thus

$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{[4(k+1)-3][4(k+1)+1]}$

$= \frac{k}{4k+1} + \frac{1}{[4(k+1)-3][4(k+1)+1]}$ by (*)

3 cont'd

$= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)}$

$= \frac{k(4k+5) + 1}{(4k+1)(4k+5)}$

$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)}$

$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)}$

$= \frac{(k+1)}{(4k+5)}$

Thus if true for $n=k$, it is also true for $n=k+1$.

$\therefore$ Since it is true for $n=1$, it is true for $n=1+1=2$, and thus also for $n=2+1=3$,

and so on for all integers $n \geq 1$.

QUESTION 4

(a) $\int \cos^2 3x \, dx = \int \left(\frac{1}{2} + \frac{1}{2} \cos 6x \right) dx$

$= \frac{1}{2} x + \frac{1}{12} \sin 6x + C$

(b) (i) Graphing $2x(1-x)$ versus x gives a

concave-down parabola

with axis of symmetry

$x = \frac{1}{2}$.

$\therefore$ maximum value is $2\left(\frac{1}{2}\right)\left(1-\frac{1}{2}\right) = \frac{1}{2}$

[Alternatively, Let $f(x) = 2x(1-x)$

$\therefore f'(x) = 2 - 4x, f''(x) = -4$.

For max., $f'(x) = 0 \therefore x = \frac{1}{2}$

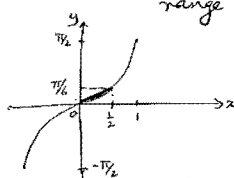
$f(x) = \frac{1}{2}$

Since $f''(x) < 0 \therefore \frac{1}{2}$ is a maximum value.]

(ii) If $0 \leq x \leq 1$ then $0 \leq 2x(1-x) \leq \frac{1}{2}$ [from (i)]

Thus $\sin^{-1}\{2x(1-x)\}$ has

range $R: 0 \leq y \leq \frac{\pi}{6}$.



✓ refers to $a > 0$ since $x > 0$

✓ correct sub. into correct formula

✓ correct answer

✓

✓ authentic simplification

✓

✓

✓ proves for $n=1$

✓ correct start

✓ correct use of trig relation

✓ correct answer

✓ a correct demonstration

✓ relevant working

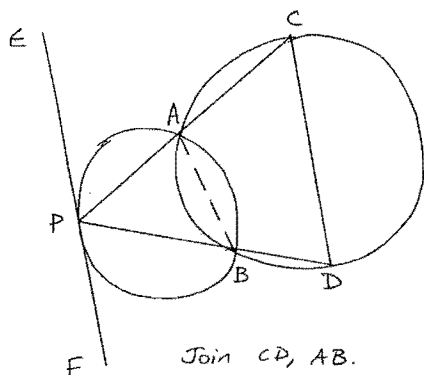
✓ correct answer

✓ correct working/simplifying

✓

Q4 cont'd

(d)



$\angle EPA = \angle ABP$ (alternate segment theorem) ✓

$\angle ABP = \angle ACD$ (exterior angle of cyclic quad. ACDB) ✓

$\therefore EF \parallel CD$ (alternate angles $\angle EPA$ and $\angle ACD$ are equal) ✓

Q4 cont'd

(c) (i) Let $T = 22 + Ae^{kt}$

Then $\frac{dT}{dt} = k \cdot Ae^{kt}$
 $= k(T - 22)$

So $T = 22 + Ae^{kt}$ is a solution.

(ii) Measure time t (in hours) from 11:00pm.

Thus at $t=0$, $T=31$

$t=1$, $T=30$.

That is, $31 = 22 + Ae^0 \therefore A = 9$

and $\therefore 30 = 22 + 9e^k$

$\therefore \frac{8}{9} = e^k$

$\therefore k = \ln\left(\frac{8}{9}\right) \left(\approx -0.117783\right)$ ✓ correct to value

$\therefore 37 = 22 + 9e^{\ln(\frac{8}{9})t}$

$\therefore \frac{15}{9} = e^{\ln(\frac{8}{9})t}$

$\therefore t = \frac{\ln(\frac{15}{9})}{\ln(\frac{8}{9})}$

$= -4.337$

$\therefore$ he was murdered at $(11 - 4.337)$ pm.

$= 6:40$ pm (nearest minute). ✓ correct answer

QUESTION 5

(a) $\int_2^4 \frac{x}{(x-1)^2} dx$. Let $u = x-1$

$\therefore du = dx$
 $x = u+1$

$x=2 \rightarrow u=1$
 $x=4 \rightarrow u=3$

$\therefore \int = \int_1^3 \frac{u+1}{u^2} du$

$= \int_1^3 \left(\frac{1}{u} + u^{-2}\right) du$

$= \left[\ln u - \frac{1}{u}\right]_1^3$

$= \left(\ln 3 - \frac{1}{3}\right) - (\ln 1 - 1)$

$= \ln 3 + \frac{2}{3}$

(≈ 1.765) . ✓ correctly set-up for u

(b) (i) $x = 2 \cos 3t$

$\therefore \dot{x} = -6 \sin 3t$

$\therefore \ddot{x} = -18 \cos 3t$

$\therefore \ddot{x} = -9x$

This is in the form $\ddot{x} = -n^2x$, hence it's S.H.M.

(ii) $T = \frac{2\pi}{3}$ seconds

(iii) When $x=1$, $1 = 2 \cos 3t$

$\therefore 3t = \frac{\pi}{3}$

$\therefore t = \frac{\pi}{9}$ seconds

At $t = \frac{\pi}{9}$, $|\dot{x}| = |-6 \sin 3(\frac{\pi}{9})| = 3\sqrt{3}$ cm/s. ✓ correct time

Q5 cont'd

(b) (iii) Alternatively, $v^2 = n^2(a^2 - x^2)$

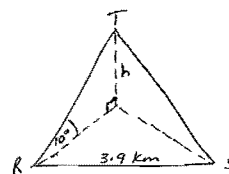
$= 9(4 - 1^2)$

$= 27$

$\therefore v = \pm\sqrt{27}$

$\therefore \text{speed} = |v| = 3\sqrt{3}$ cm/s ✓ correct use of formula

(c)



$\angle RTS = 180^\circ - (65^\circ + 48^\circ)$
 $= 67^\circ$

$\therefore \frac{TR}{\sin 48^\circ} = \frac{3.9}{\sin 67^\circ}$

$\therefore TR = \frac{3.9 \sin 48^\circ}{\sin 67^\circ}$

But $\frac{h}{TR} = \sin 10^\circ$

$\therefore h = \frac{3.9 \sin 48^\circ}{\sin 67^\circ} \cdot \sin 10^\circ$

≈ 0.5467 km

$= 547$ m. (nearest metre). ✓ correct expression for TR

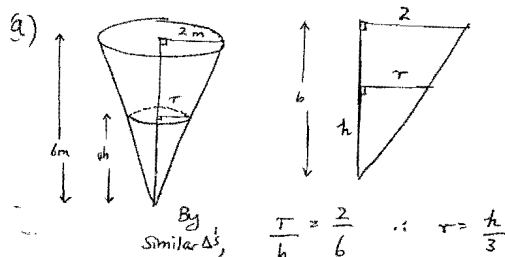
✓ Finds $\angle R$

✓ correct expression for h

✓ correct expression for h

✓ correct answer

QUESTIONS 6



$$\text{Thus } V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi \left(\frac{h}{3}\right)^2 h$$

$$\therefore V = \frac{\pi h^3}{27}$$

Now $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$, (where $\frac{dV}{dh} = \frac{\pi h^2}{9}$)

$$\therefore 300000 \frac{\text{cm}^3}{\text{min}} = \frac{\pi}{9} (200)^2 \text{ cm}^2 \cdot \frac{dh}{dt}$$

$$\therefore \frac{dh}{dt} = \frac{9}{\pi (200)^2} \cdot 300000 \frac{\text{cm}}{\text{min}}$$

$$= 21.4859$$

$$= 21.5 \text{ cm/min. (1 d.p.)}$$

✓ expresses r in terms of h

✓ correct expression for V

✓ correct use of Chain Rule

✓ correct answer

Q6 cont'd

(b) (i) $y = \frac{x^2}{4a} \therefore \frac{dy}{dx} = \frac{x}{2a}$

$$\therefore \text{at } P, \frac{dy}{dx} = \frac{2ap}{2a} = p$$

$$\therefore \text{tangent is } y - ap^2 = p(x - 2ap)$$

$$\therefore y = px - ap^2$$

At $T, y=0 \therefore px - ap^2 = 0$

$$x = \frac{ap^2}{p}$$

$$\therefore x = ap$$

$$\text{Hence } T = (ap, 0)$$

(ii) $S(0, a), T(ap, 0)$, let $k:l = 2:-1$

$$\text{Then } M = \left[\frac{(2)(ap) + (-1)(0)}{2 + (-1)}, \frac{2(0) + (-1)(a)}{2 + (-1)} \right]$$

$$\therefore M = (2ap, -a)$$

(iii) As P moves on the parabola, M moves along the line $y = -a$;
 M moves along the directrix of the parabola.

(c) Let $y = f(x)$, i.e. $y = \log_e \left(\frac{1+x}{1-x} \right)$

For inverse, $x = \log_e \left(\frac{1+y}{1-y} \right)$

$$\therefore e^x = \frac{1+y}{1-y}$$

$$e^x - e^x y = 1 + y$$

$$\therefore e^x - 1 = y(e^x + 1)$$

$$\text{Hence } y = \frac{e^x - 1}{e^x + 1}$$

✓ derives $\frac{dy}{dx}$

✓ derives equation of tangent

✓ shows $T = (ap, 0)$

✓ correct use of division formula

✓ correct coordinates of M

✓ geometric answer.

✓ correct to here

✓ correct answer

QUESTION 7

i) (i) $\ddot{x} = 0$
 $\dot{x} = c$
 at $t=0, \dot{x} = V \cos \theta$
 $\therefore \dot{x} = V \cos \theta$
 $\therefore x = Vt \cos \theta + c$
 at $t=0, x=0 \therefore c=0$
 $\therefore x = Vt \cos \theta$

$\ddot{y} = -g$
 $\dot{y} = -gt + c$
 at $t=0, \dot{y} = V \sin \theta$
 $\therefore c = V \sin \theta$
 $\therefore \dot{y} = -gt + V \sin \theta$
 $\therefore y = -\frac{gt^2}{2} + Vt \sin \theta + c$
 at $t=0, y=1.8 \therefore c=1.8$
 $\therefore y = -\frac{gt^2}{2} + Vt \sin \theta + 1.8$

Correct derivation + use of initial conditions

✓✓

✓ finds t

✓ correct substitution + simplifying

✓ correct equations for acorn

✓ writes h in terms of d

✓ correct conclusion

(ii) "Arrow reaches tree" means $x=d$
 $\therefore d = Vt \cos \theta \therefore t = \frac{d}{V \cos \theta}$
 $\therefore$ at this time,
 $y = -\frac{g}{2} \left(\frac{d}{V \cos \theta} \right)^2 + V \left(\frac{d}{V \cos \theta} \right) \sin \theta + 1.8$
 i.e. $H = -\frac{gd^2 \sec^2 \theta}{2V^2} + d \tan \theta + 1.8$

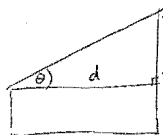
(iii) The acorn's equations of motion are
 $x=d, y=h - \frac{gt^2}{2}$

Thus at $t = \frac{d}{V \cos \theta}, y = h - \frac{g}{2} \left(\frac{d}{V \cos \theta} \right)^2$
 $y = h - \frac{gd^2 \sec^2 \theta}{2V^2}$

But $\frac{h-1.8}{d} = \tan \theta \therefore h = d \tan \theta + 1.8$

Hence $y = d \tan \theta + 1.8 - \frac{gd^2 \sec^2 \theta}{2V^2} = H$

i.e. the acorn is the same height above the ground as the arrow arrives at the tree.



Q7 cont'd

(b) John will succeed if he throws a total of 5 on the next toss, or should he not throw a 5 or 7 on this toss but throws 5 on the next; or should he not throw a 5 or 7 on either of these tosses but throw 5 on the next; etc.

$$\therefore P(\text{throws 5 before 7})$$

$$= \frac{4}{36} + \frac{26}{36} \cdot \frac{4}{36} + \left(\frac{26}{36} \right)^2 \cdot \frac{4}{36} + \dots$$

$$= \frac{\frac{4}{36}}{1 - \frac{26}{36}} \quad \left(\text{since } S_{\infty} = \frac{a}{1-r} \right)$$

$$= \frac{2}{5}$$

✓ correct logic + series

✓ correct answer

[Alternatively,

John keeps throwing until he gets either a 5 or a 7. We need to find the probability the 5 comes up before the 7.

So the question is, given he gets a 5 or a 7, what is the probability that it is a 5?

There are 10 ways to throw a 5 or a 7, and of these four give a 5.

So the probability he gets a 5 before a 7 is just $\frac{4}{10} = \frac{2}{5}$.

✓✓

(no marks for just finding $P(5 \text{ or } 7)$ on a single throw).

CONTINUED
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